

ABSTRACT

**of the PhD thesis by Yesmukhamedov Nurmaganbet Seitkaliuly on
«Information system for automatic detection of retinal diseases
based on deep learning using optical coherence tomography images»,
submitted for the degree of Doctor of Philosophy (PhD) in the
educational programme 8D06102 — Computer and Software Engineering**

General characteristics of the work. The dissertation investigates the integration of fundus image enhancement (preprocessing) with convolutional-neural-network (CNN) classification for automated diagnosis of diabetic retinopathy (DR): preprocessing is treated as an integral component of the diagnostic model rather than ancillary data preparation, because it defines the feature space available to the network.

Relevance of the research. Diabetic retinopathy is a microvascular complication of diabetes and one of the leading causes of preventable vision loss among adults of working age. Its earliest stages are asymptomatic, so detection depends not on presentation but on scheduled imaging of a cohort defined by a systemic diagnosis: screening is, above all, a problem of volume, and manual grading scales linearly with the finite and geographically concentrated time of specialists. Automation has been technically feasible for a decade, so the question lies in the conditions under which the demonstrations were obtained: photographs acquired by different cameras, operators and dilation states vary systematically, and where screening capacity is scarcest, acquisition conditions are least controlled. The transform applied before the first convolution determines the network's feature space, so a model reported without its preprocessing specified has not been fully described, and comparisons between such models are comparisons between partly unknown systems.

The aim of the research is to develop and experimentally validate an integrated fundus image enhancement and CNN classification framework for automated five-class DR diagnosis, in which an 8-stage preprocessing pipeline is specified as an integral component of the model, and to establish, under controlled contrast against a configuration trained without it, what difference that specification makes to performance, transferability and interpretability.

Research objectives: 1. To analyse the problem domain — the grading of the disease, the screening requirements of resource-limited settings, the sources of image-quality loss — and to formulate the research problem (Chapter 1). 2. To specify the methodology: the pipeline, the classification architectures, the pretraining and fine-tuning strategy, the explainability formalism and the evaluation protocol (Chapter 2). 3. To evaluate the integrated configuration in domain and on seven external corpora and to validate the results statistically (Chapter 3). 4. To build a screening system suited to settings without inference acceleration and with intermittent connectivity (Chapter 4).

The object of the research is the process of automated multi-stage diagnosis of diabetic retinopathy from colour fundus photographs by means of convolutional networks.

The subject of the research is the set of methods by which preprocessing is integrated with CNN classification, together with the properties of the resulting configuration: five-class diagnostic performance, transferability to corpora not seen in training, alignment of attention with expert annotation, and behaviour across camera domains.

Research methods. The methodological framework is controlled experimental comparison, with everything outside the manipulated factor held fixed. Its instruments: optic-disc and fovea localisation by a frozen detector; isotropic rescaling with padding made explicit to the network as a mask channel; adaptive illumination correction and dual-constraint CLAHE in the LAB colour space; ResNet-50 and EfficientNet-B3 under a weighted focal loss; Grad-CAM with the Attention–Lesion Overlap and IoU measures; maximum mean discrepancy over penultimate-layer features; and patient-level stratified 5-fold cross-validation.

The empirical basis of the research is eight public and clinical fundus datasets from cameras of four manufacturers (Canon, Topcon, Kowa, Zeiss): EyePACS (~35,126 graded images, training and ablation) and its unlabelled split (~53,576), disjoint from the training set, for in-domain pretraining; APTOS 2019 (~3,662, cross-dataset transfer); IDRiD (516, of which 81 with lesion masks) and Messidor-2 (~1,748) for external clinical evaluation; DDR (~13,673), ODIR-5K (~5,000) and RFMiD (~3,200) for device shift; and a Kazakh clinical dataset (60 images, 30 patients $\times$ 2 eyes) for the data-scarce regime. Results are reported per corpus and never aggregated.

The scientific novelty of the research consists in the following: 1. The principal contribution is conceptual: preprocessing is formalised as a binding part of the model specification and placed under controlled contrast, so that what is new is not the existence of a pipeline but the treatment of one as an object of experiment. 2. Five elements of its engineering realisation are specified in forms not previously combined: isotropic resize with centred padding; a field-of-view mask as a fourth input channel; illumination correction scaled to per-image geometry and applied inside the mask only; channel statistics from valid fundus pixels; and canonical orientation whose augmentation dispersion derives from landmark localisation uncertainty. 3. The clip limit of the contrast stage is the minimum of a histogram-relative and a tile-relative constraint, applied stochastically so the stage serves as both enhancement and regularisation. 4. Attention alignment is measured by the fraction of an annotated lesion that model attention covers. 5. The postulated mechanism is measured directly: source-to-target distance is computed at the penultimate layer under both configurations across six external corpora. 6. A defect shared by measures in common use for external robustness is identified: each normalises an arm's external performance against that same arm's in-domain performance.

The main results of the research. 1. An 8-stage preprocessing pipeline was designed and formalised: canonical flip, OD-fovea rotation normalisation, field-of-view crop with isotropic resize, FOV mask as a fourth input channel, illumination

correction, stochastic CLAHE, augmentation and dataset-specific normalisation. 2. The experimental programme was carried out, its outcomes stated in the provisions submitted for defence. 3. A modular architecture for a DR screening system in resource-limited settings was designed and a working demonstrator of it was built.

Main provisions put forward for defence: 1. Preprocessing of fundus images is a formalisable and experimentally testable component of the diagnostic model rather than ancillary preparation of the data; the provision is methodological. 2. The integrated configuration exceeds the baseline on the training corpus of 35,126 images, on both architectures, in all three components of the conjunctive criterion: weighted F1 by 6.54 and 6.55 percentage points, area under the curve by 0.032 and 0.036, and quadratic kappa by 0.11; every interval excludes zero, the comparisons survive correction for multiplicity, and there is no interaction between arm and architecture. The arms differ in initialisation as well, so the effect belongs to the configuration as a whole. 3. The contributions of the individual stages are separable: cumulative ablation over eight levels raises weighted F1 from 0.7538 to 0.8193, monotonically and without a single inversion in any of the five folds, each of the seven transitions contributing 0.0065 to 0.0143, the two photometric stages carrying 41% of the gain, and both parameters exhibiting an interior optimum confirmed on held-out data: a clip factor of 2.5 with a histogram threshold of 0.03, and a correction scale of 0.07 of the field diameter. 4. Distance from the training distribution falls on every one of six external corpora — by 0.070 to 0.093 at the penultimate layer, with every interval excluding zero — without the transform observing any target statistics. 5. Competence transfers to corpora not seen in training: on the cross-corpus set a generalisation ratio of 0.898 against 0.858 and a weighted F1 higher by 0.089; across five camera groupings performance is higher on every one and the spread contracts by factors of 2.4 and 3.1; and on two external clinical corpora the margin is 0.069 and 0.054 in weighted F1, above the minimal clinically important difference of 0.050. 6. Model attention overlaps expert-annotated lesions more closely on all four annotated lesion types — by 0.099 to 0.129 at $p \leq 0.0148$ — and the direction holds at every binarisation threshold; the provision is defended as alignment, not as localisation of pathology. 7. Also submitted are a coupled thermal-optical model of fundus tissue response under laser exposure as a theoretical contribution and a modular screening-system architecture as a design one.

Theoretical significance lies in how the problem is posed and measured: the reframing changes what counts as a complete description of a diagnostic model of this class and as a fair comparison; five formalisations make previously informal choices explicit and testable, and a mechanism hitherto invoked as an explanation is made measurable at the penultimate layer, which makes the mechanistic claim falsifiable independently of the performance claims.

Practical significance. Four things are made practically available, each with its limit: a fully specified preprocessing regime whose parameter values should be re-established rather than inherited; a screening system with preprocessing, inference, store-and-forward telemedicine and physician-in-the-loop decision-support modules, integrable with PACS/EHR systems and the national eHealth platform — its demonstrator is deployed and performs inference, while the remaining parts stay

design specifications and the data-protection provisions are alignment by design rather than certified compliance; a protocol for ingesting externally sourced images, validated only against the clinical source it was built for; and an argument of applicability to the national screening context, bounded by the absence of field testing there. The clinician retains the diagnosis and the responsibility for it.

Reliability of the results rests on procedure rather than on magnitude: patient-level grade-stratified partitioning, a hierarchy of measures fixed in advance, paired testing on identical cases, resampling, and a mixed-effects model. The acceptance criterion for each hypothesis was fixed before the experiment that tested it. Three qualifications bound the apparatus in general: correction for multiplicity is scoped to the single experiment within which the comparisons were planned; several evaluations rest on the models of one fold and understate total uncertainty; and one experiment depends on a clinical corpus that cannot be redistributed. Four attach to particular provisions: the stage ranking holds at the resolution of groupings only, and the two parameter optima are properties of this corpus; the reduction in distance holds in direction only; neither the generalisation ratio nor the camera-group floor discriminates between the arms, and on the second external clinical corpus the margin over the minimal important difference is four thousandths; and the attention result rests on one annotated corpus. Not submitted for defence are clinical validity, readiness for unsupervised deployment, device certification, localisation of pathology, and superiority over any named published system.

Approbation of the results and connection with scientific programmes. The results were reported and discussed at the 3rd International Workshop on Digital Society (DS 2025, Istanbul, Türkiye, 28–30 October 2025). The direction of the research corresponds to the state priorities of the Republic of Kazakhstan in the digitalisation of healthcare and the development of artificial intelligence, in particular the Concept for the Development of Artificial Intelligence for 2024–2029 and the Law of the Republic of Kazakhstan «On Artificial Intelligence» of 17 November 2025. The work is carried out in accordance with subparagraph 2 of paragraph 3 of article 20 of the Law of the Republic of Kazakhstan «On Science». This is a correspondence between the direction of the research and published national priorities, not a statement of funding under a state programme.

Publications. The main results of the dissertation are published in 5 scientific works: 1 article in a journal indexed in Scopus and Web of Science (*Eastern-European Journal of Enterprise Technologies*, Q3, CiteScore 2025 = 2.5, 42nd percentile in the category Computer Science Applications); 1 paper in Scopus-indexed international conference proceedings (*Procedia Computer Science*); and 3 articles in journals recommended by the Committee for Quality Assurance in Science and Higher Education of the Republic of Kazakhstan. The full record is given in the list of published works on the dissertation topic. In addition, the software complex implementing the method is entered in the State Register of Rights to Objects Protected by Copyright of the Republic of Kazakhstan (certificate No. 78109 of 4 September 2026); it is reproduced in Appendix F.

Main content of the work. The introduction substantiates the relevance of the topic and formulates the aim, the objectives, the object, the subject, the novelty, the

provisions submitted for defence, the significance, the reliability and the approbation. Chapter 1 establishes the clinical and technical context of screening and formulates the research problem; Chapter 2 specifies the methodology — the pipeline, the classification architectures and their adaptation to a four-channel input, the pretraining strategy, the explainability formalism and the evaluation protocol fixed in advance; Chapter 3 reports the experimental programme — the factorial contrast, the stage ablation, the measurement of domain distance, cross-corpus, device and clinical transfer, attention alignment and the data-scarce regime; Chapter 4 describes the screening system, distinguishing what exists from what remains specification. The conclusion consolidates the outcomes and the directions of further work.

Author's personal contribution. The principal results were obtained by the author personally: the reframing of preprocessing as an integral component of the diagnostic model; the design and formal specification of the 8-stage pipeline; the Attention–Lesion Overlap measure; the experimental design and the statistical validation framework; and the architecture of the screening system. All five publications are co-authored; in each the author contributed the part on which this dissertation rests.

Structure and scope of the work. The dissertation comprises front matter, an introduction, four chapters, a conclusion, a list of references and six appendices. The dissertation is set out on 113 pages and contains 19 tables and 16 figures. The list of references comprises 102 sources. The counts are of the dissertation excluding the appendices.